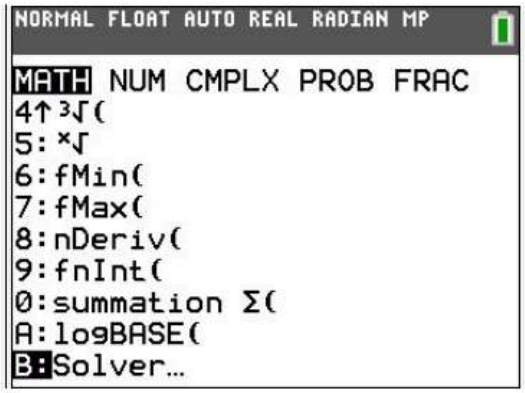
TI-84 Calculator fnInt Feature



Numerical Integration on TI-84:
Press [MATH]
Select fnInt
Input the function, variable, and lower/upper bounds

Example: To find $\int_0^2 x^2 \cdot dx$

Input fnInt(X^2, X, 0, 2). for older TI-84

Input $\int_0^2 (x^2)dx$ for newer TI-84

1) Find the length of the arc for $r = 4\cos\theta \qquad 0 \le \theta \le \pi$

a) Graph $r = 4\cos\theta \qquad 0 \le \theta \le \pi$

(multiple choice). Choose the correct graph (blue, red, green, or orange)

b) $\dfrac{dr}{d\theta} =$ ___?___

c) Arc Length $= \displaystyle\int_{\alpha}^{\beta} \sqrt{f[(\theta)]^2 + [f'(\theta)]^2}\, d\theta$

$$= \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta = \underline{\quad ? \quad}$$

Hint: Use TI-84 fnInt feature to evaluate the definite integral.

Answers

a) b) c)

2) Find the length of the arc for $r = 4 + 2\cos\theta \quad 0 \le \theta \le \pi$

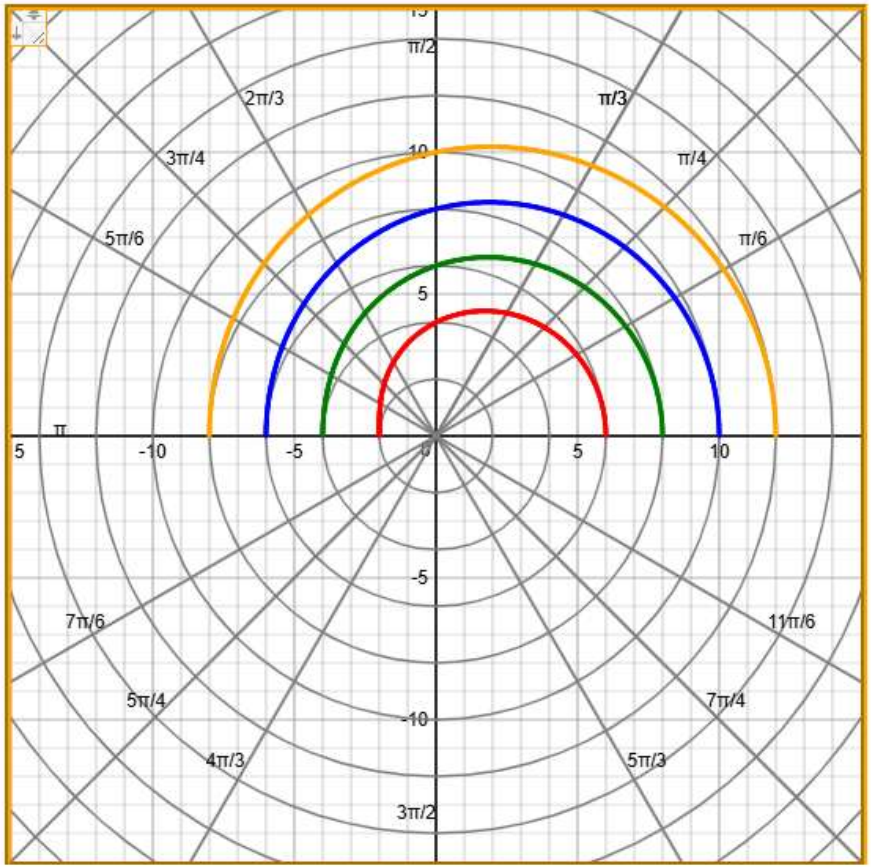
a) Graph $r = 4 + 2\cos\theta \quad 0 \le \theta \le \pi$

b) $\dfrac{dr}{d\theta} =$ ___?___

c) Arc Length $= \displaystyle\int_{\alpha}^{\beta} \sqrt{f[(\theta)]^2 + [f'(\theta)]^2}\, d\theta$

$$= \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta = \underline{\quad ? \quad}$$

Hint: Use TI-84 fnInt feature to evaluate the definite integral.



Answers

a) b) c)

3) Find the length of the arc for $r = 4 \qquad 0 \le \theta \le 2\pi$

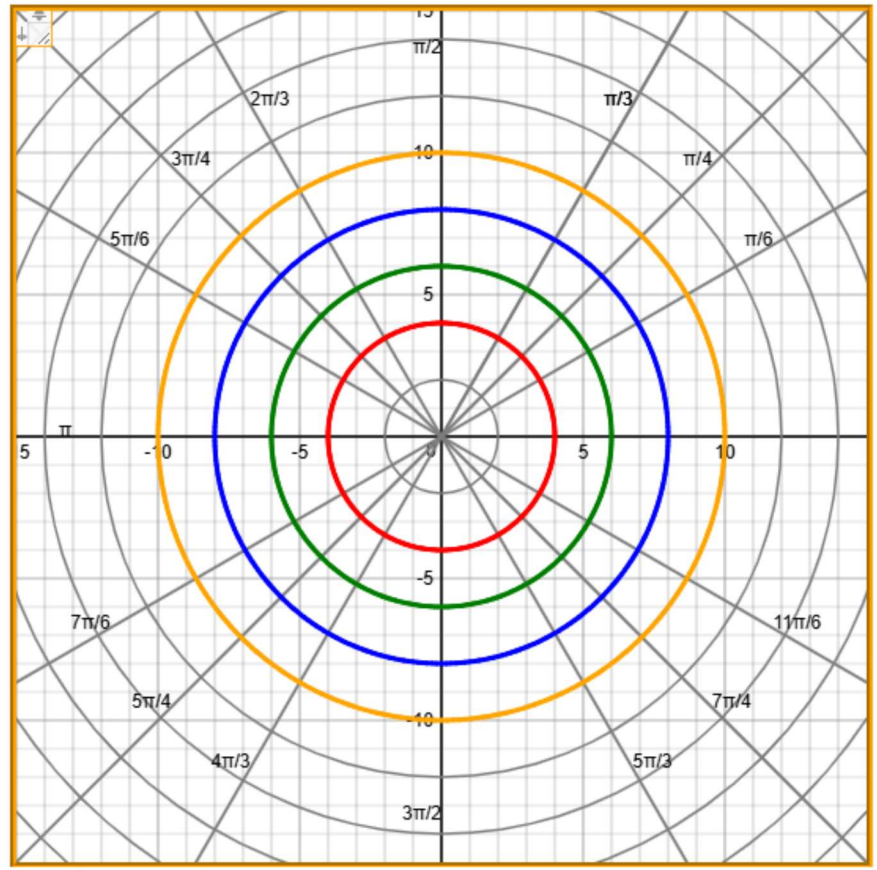
a) Graph $r = 4 \qquad 0 \le \theta \le 2\pi$

b) $\dfrac{dr}{d\theta} =$?

c) Arc Length $= \displaystyle\int_{\alpha}^{\beta} \sqrt{f[(\theta)]^2 + [f'(\theta)]^2}\, d\theta$

$$= \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta = \text{?}$$

Hint: Use TI-84 fnInt feature to evaluate the definite integral.



Answers

a) b) c)